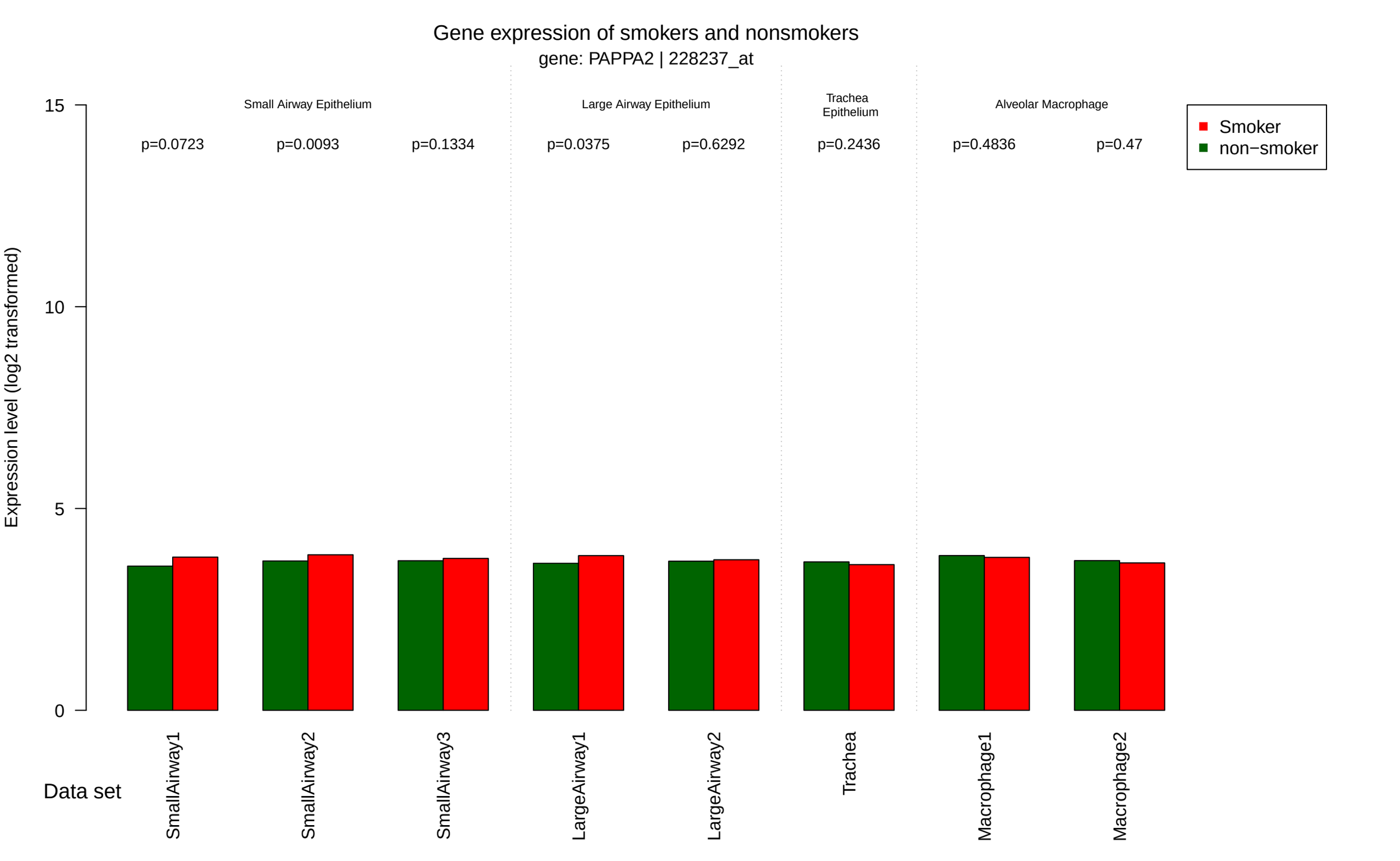
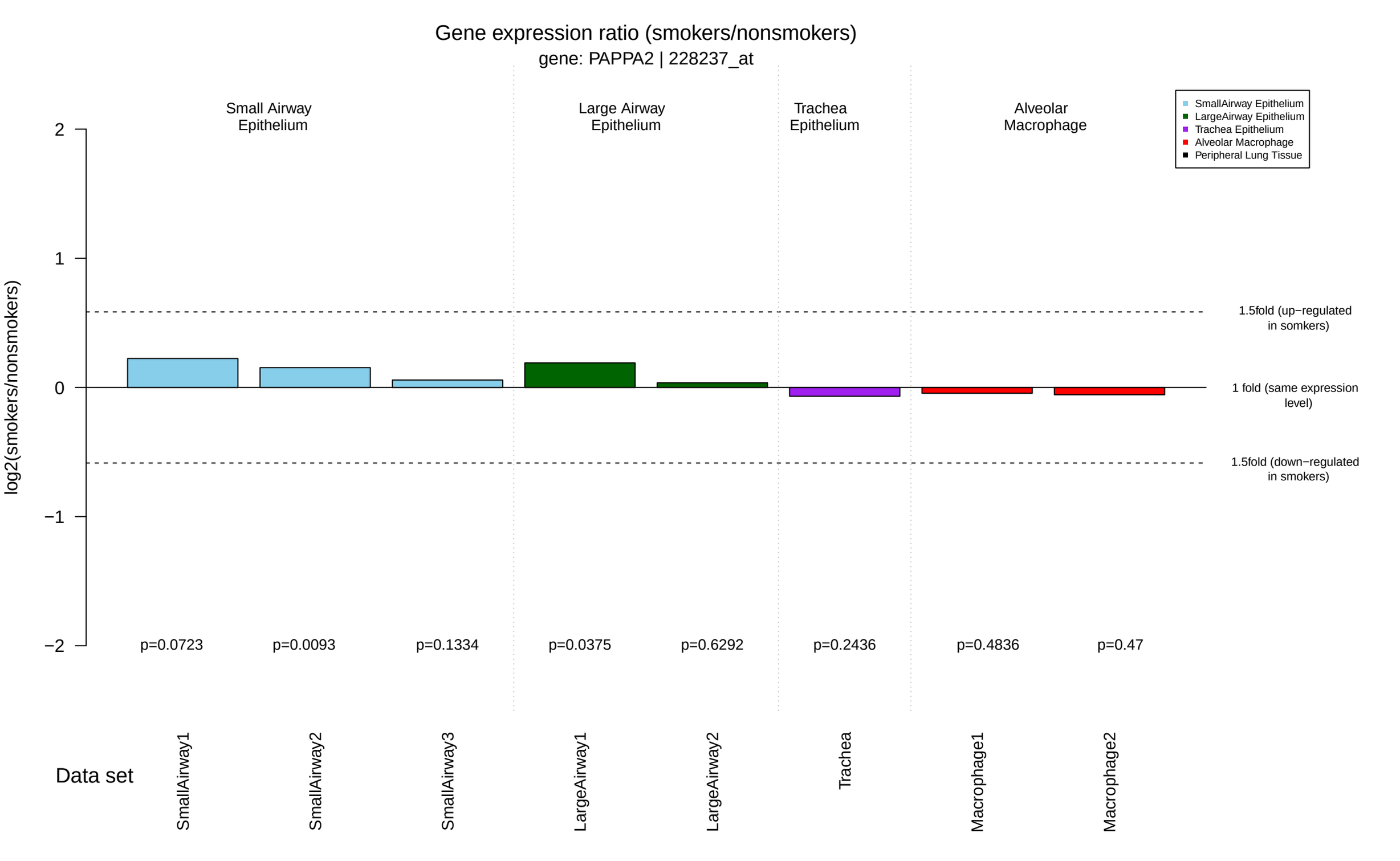
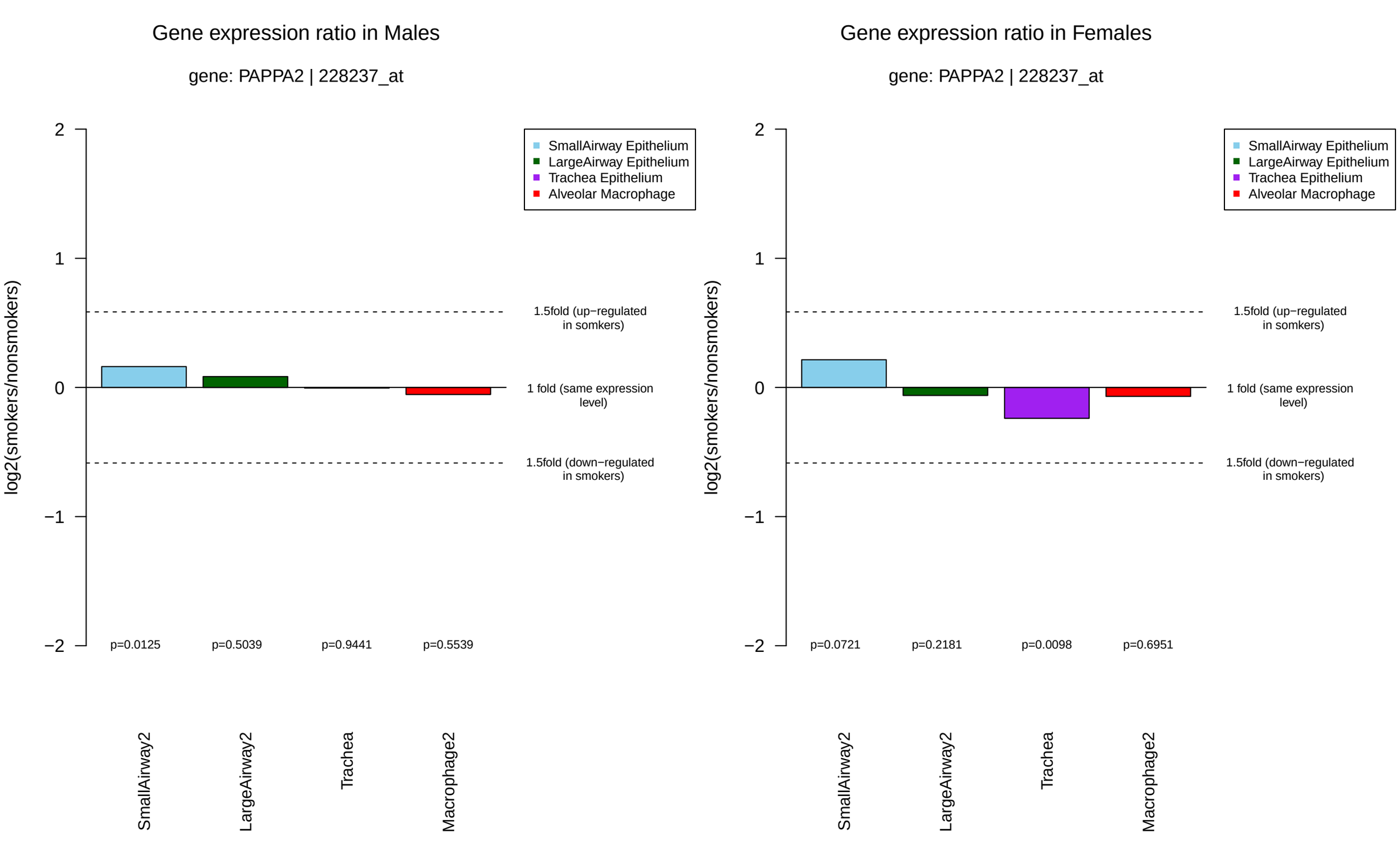
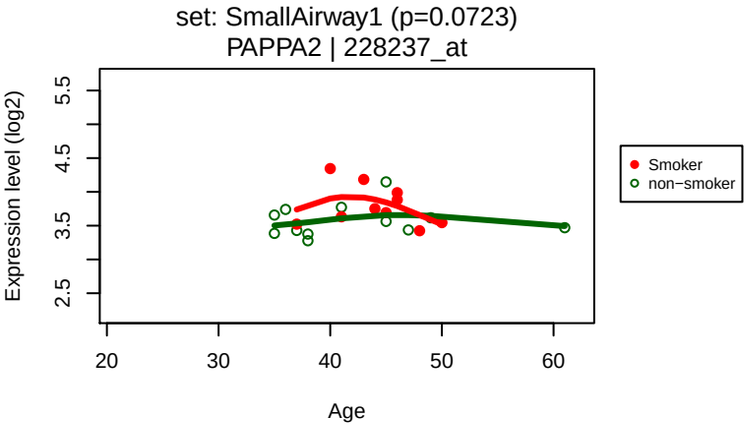
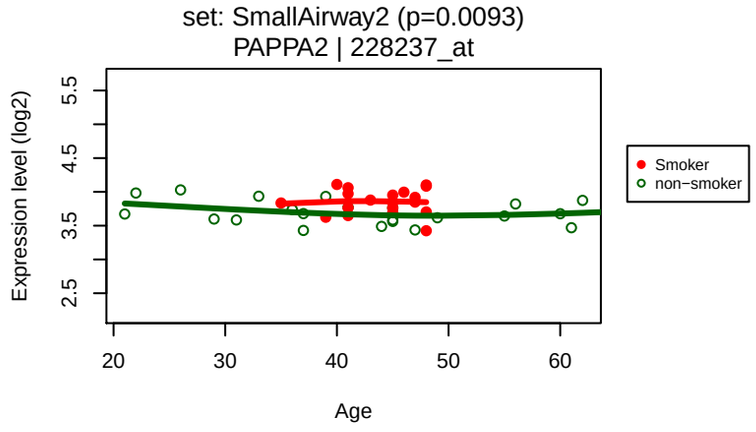
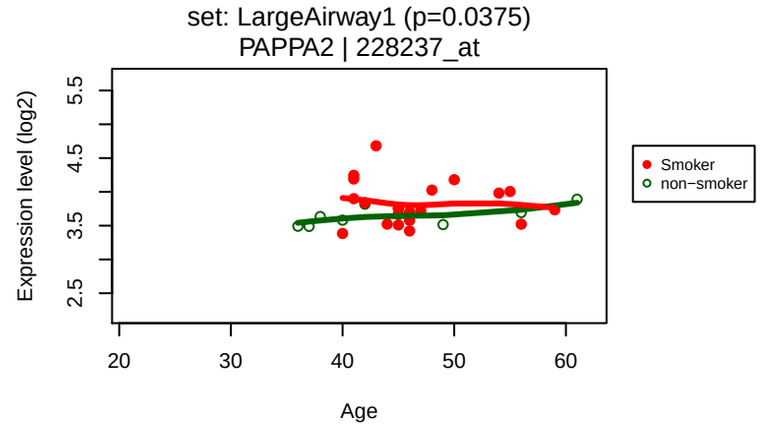

Gene expression of smokers and nonsmokers
gene: PAPPA2 | 228237_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0723
p=0.0093
p=0.1334
p=0.0375
p=0.6292
p=0.2436
p=0.4836
p=0.47
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

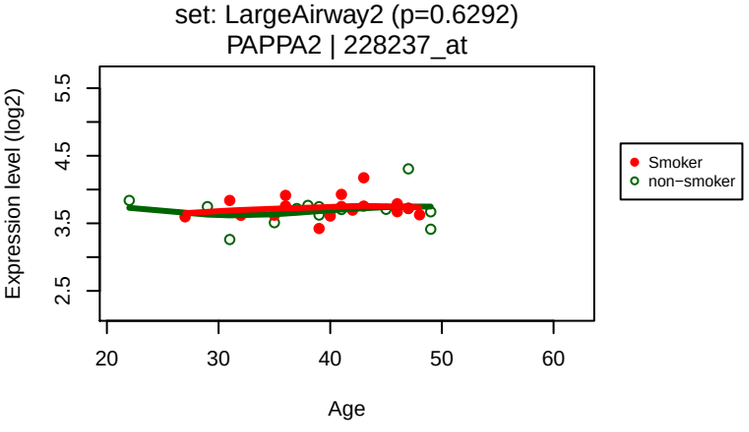

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA2 | 228237_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0723
p=0.0093
p=0.1334
p=0.0375
p=0.6292
p=0.2436
p=0.4836
p=0.47
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

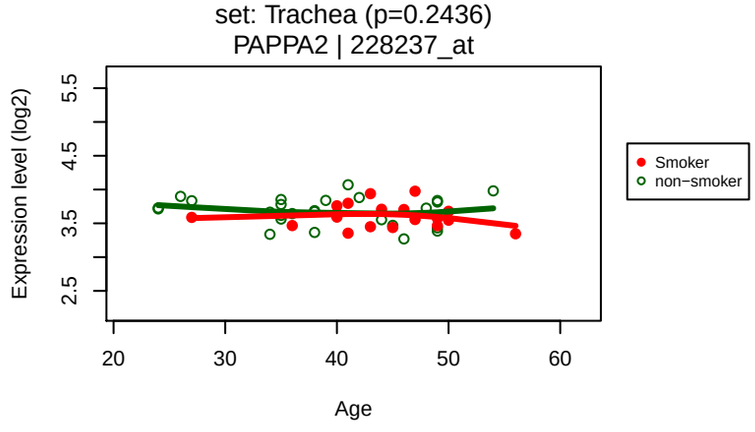

Gene expression ratio in Males
gene: PAPPA2 | 228237_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0125
p=0.5039
p=0.9441
p=0.5539
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA2 | 228237_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0721
p=0.2181
p=0.0098
p=0.6951
SmallAirway2
LargeAirway2
Trachea
Macrophage2

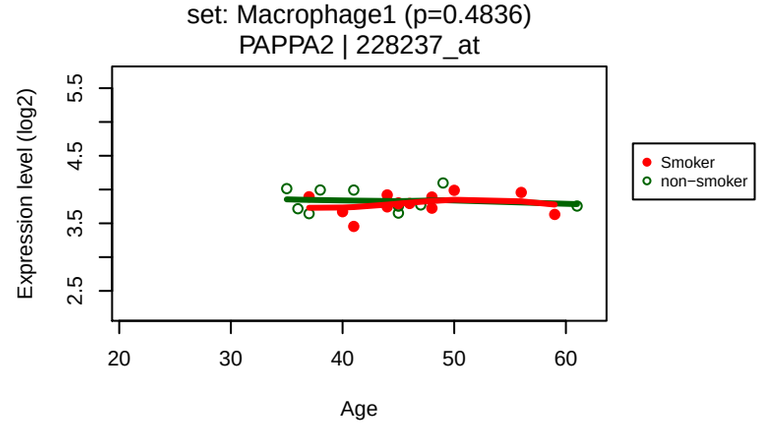

set: SmallAirway1 (p=0.0723)
PAPPA2 | 228237_at
Expression level (log2)
Age
Smoker
non−smoker

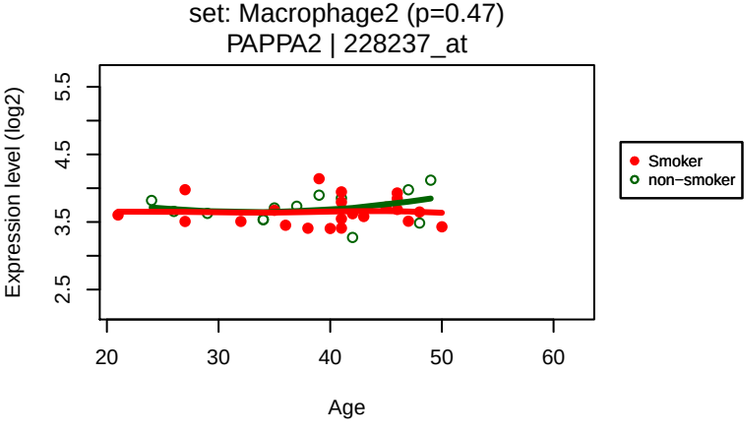

set: SmallAirway2 (p=0.0093)
PAPPA2 | 228237_at
Expression level (log2)
Age
Smoker
non−smoker

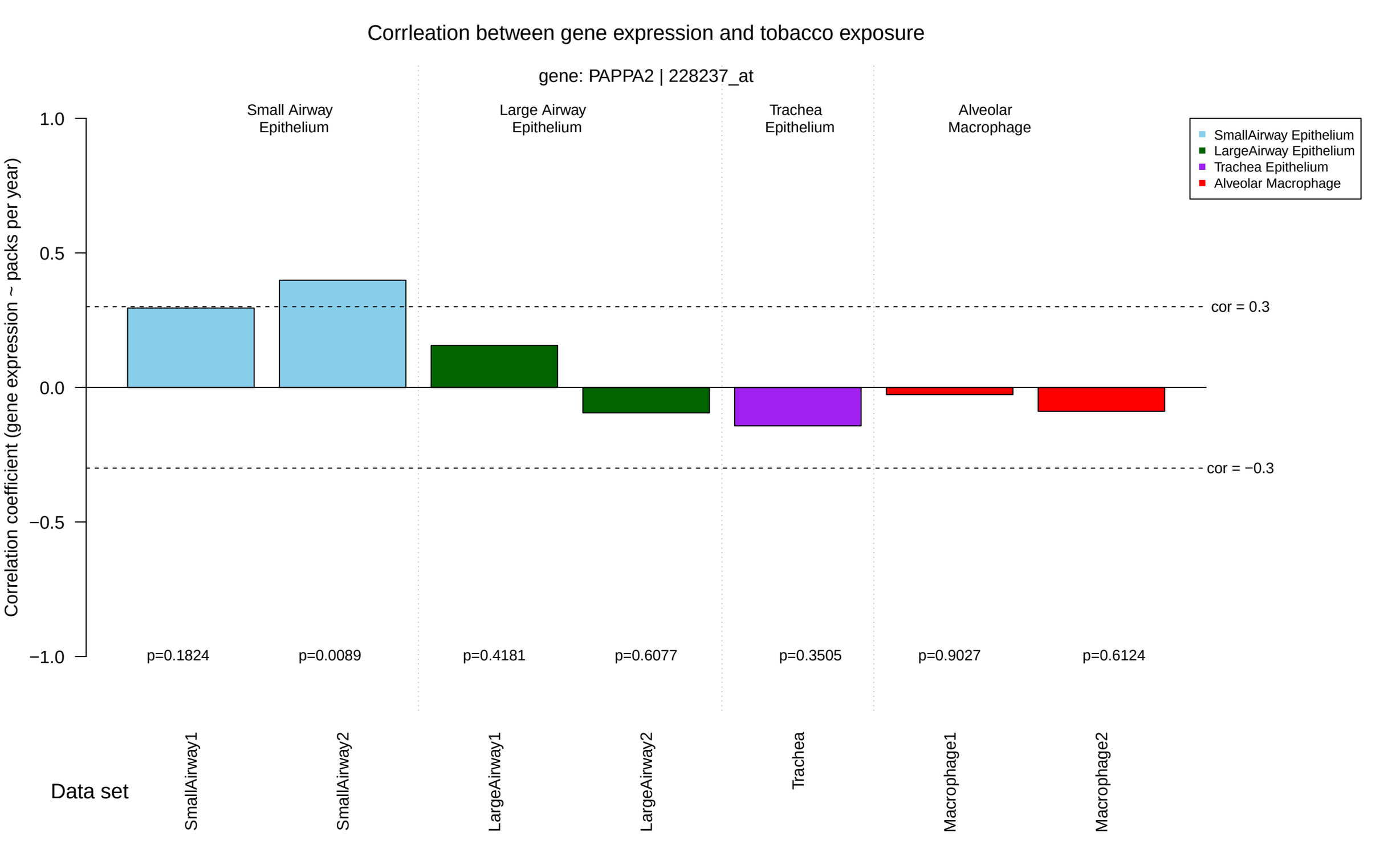

set: LargeAirway1 (p=0.0375)
PAPPA2 | 228237_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6292)
PAPPA2 | 228237_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2436)
PAPPA2 | 228237_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4836)
PAPPA2 | 228237_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.47)
PAPPA2 | 228237_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA2 | 228237_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1824
p=0.0089
p=0.4181
p=0.6077
p=0.3505
p=0.9027
p=0.6124
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2